

ISyE6669 Deterministic Optimization
Fall 2017
Final Exam Practice

1. Answer the following questions.

- (a) True or False (give a short justification of your answer): Consider a feasible integer optimization (minimization) problem and its Lagrangian dual problem. It is possible that the Lagrangian dual problem's optimal objective value is strictly greater than the integer program's optimal objective value.

Solution: False. Lagrangian dual is a relaxation.

- (b) True or False (give a concise justification): Consider the linear programming relaxation of an integer optimization (minimization) problem. If the optimal solution of the linear relaxation problem is integral, then it is also the optimal solution of the integer program.

Solution: True.

- (c) True or False (give a concise justification): Consider the LP relaxation and the Lagrangian dual of an integer optimization (minimization) problem. It is possible that the optimal cost of the LP relaxation is strictly larger than the optimal cost of the Lagrangian dual.

Solution: False. The Lagrangian dual is at least as good as the LP relaxation in terms of the lower bound to the optimal cost of the IP problem, but could be strictly better than the LP relaxation in terms of lower bound.

- (d) True or False (give a short justification of your answer): Using branch-and-bound method to solve a maximization integer program, suppose the current best integer solution has an objective value 23. At a branch-and-bound node, the linear programming relaxation gives a fractional solution with objective value 23. We should stop branching at this node.

Solution: True. Because any children nodes from this node will have optimal value lower than or equal to 23. But we already have an integer feasible solution with an objective value 23, so there is no need to further branch at this node.

- (e) Is the following problem a convex optimization problem? If yes, you need to show the convexity of the objective function and the feasible region. Otherwise, reformulate the problem as a mixed-integer programming problem with linear objective function and linear constraints. If you need to introduce other variables or big-M constant, define them clearly.

$$\begin{array}{ll}\min & x + 2y \\ \text{s.t.} & |x - y| \geq 1 \\ & 0 \leq x \leq 3 \\ & 0 \leq y \leq 2\end{array}$$

Solution: The remaining solutions, see the attached scan files.

(f) Consider the set

$$S = \{(x_1, x_2, x_3, x_4, x_5) : 1.5x_1 - 0.3x_2 + 6.2x_3 + 0.9x_4 - 3.2x_5 \leq 5.2, x_1, \dots, x_5 \geq 0, \text{ integer}\}.$$

Derive a Chvatal-Gomory rounding cut.

(g) Consider a two-stage stochastic linear program:

$$\begin{aligned} \min \quad & \mathbf{c}^\top \mathbf{x} + Q(\mathbf{x}) \\ \text{s.t.} \quad & \mathbf{Ax} \leq \mathbf{b} \end{aligned}$$

where the value function $Q(\mathbf{x})$ is given by the following second-stage problems

$$Q(\mathbf{x}) = \sum_{s \in S} \min_{\mathbf{y}^s} \{(\mathbf{d}^s)^\top \mathbf{y}^s : \mathbf{T}^s \mathbf{y}^s \leq \mathbf{h}^s - \mathbf{W}^s \mathbf{x}\}.$$

Assume the second-stage problems are feasible for any $\mathbf{x}$.

- i. Show that $Q(\mathbf{x})$ is a convex piecewise linear function.
- ii. Explain why the Benders cuts generated by the Benders decomposition provide a lower estimate to $Q(\mathbf{x})$.

2. Integer Programming Modeling

John and Jennifer, as a couple, love to watch DVDs. They have decided to watch 20 DVDs together during the coming winter break. They have pre-selected 100 DVDs that they would like to watch and now want to decide which 20 out of these 100 to watch. These 100 DVDs are divided as follows: DVDs 1-20 are romantic comedies, DVDs 21-40 are action movies, DVDs 41-60 are TV series, DVDs 61-80 are science fictions, and DVDs 81-100 are documentaries.

They associated an estimated common satisfaction index s_i to each DVD $i \in \{1, \dots, 100\}$ and wish to maximize their estimated common satisfaction of watching the 20 selected DVDs during the holiday season. They also established the following constraints so that each of them will not be too unhappy: They must choose at least 5 romantic comedies and at least 5 action movies, but no more than 3 science fictions and no more than 4 documentaries. (Assume that they will be able to obtain all movies they request.)

- [illegible]

- ii. They can either watch exactly 6 DVDs out of all the romantic comedies or 3 DVDs out of DVDs 50-60.

- iii. DVDs 41-46 are seasons 1-6 of The Big Bang Theory, so they have decided that if they watch one season of The Big Bang Theory, they must watch all seasons before it. For example, if they watch season 4, they must also watch seasons 1-3.

3. Column Generation

Chemco has the following problem: One unit of raw material is used to produce 2,000g of chemical 1 and 3,000g of chemical 2. Each of these chemicals have to be immediately used to produce integral amounts of products A, B or C. Producing one unit of product A requires 140g of chemical 1 and 350g of chemical 2. Producing one unit of product B requires 250g of chemical 1 and 100g of chemical 2. Producing one unit of product C requires 125g of chemical 1 and 200g of chemical 2.

Each day, Chemco has a demand of 200 units of product A, 250 units of product B, and 300 units of product C.

Define a pattern to be the number of units of products A, B and C that will be produced using the chemicals obtained by processing 1 unit of raw material. The only restriction that each pattern must satisfy is that the total amount of chemicals 1 and 2 used does not exceed the amount obtained by processing one unit of the raw material. Here are some examples of patterns: Pattern 1 is to produce 1 unit of product A, 6 units of product B and 2 units of product C; Pattern 2 is to produce 8 units of product A, 0 units of pattern B and 1 unit of product C. Clearly, there are many more feasible patterns.

Chemco wants to minimize the number of raw material units that it uses to satisfy the demands of products A, B and C. For this purpose, it formulated an LP, where each variable x_i is associated to the i -th feasible pattern. The value of each variable represents the number of units of raw material that will be used according to each pattern. The only constraints in this LP is that all daily demands must be satisfied. However, there are too many patterns, thus too many variables in the problem, so Chemco wants to solve this problem by column generation.

Note: Chemco does not care about the variables being fractional, since it can just round the fractional values to obtain a reasonable solution.

- (a) Write down the restricted master problem with only two variables corresponding to Patterns 1 and 2 given above.

- (b) Write down the column generation subproblem.

4. Dantzig Wolfe Decomposition

$$\begin{aligned}
 (S) \quad & \min \quad x_1 + 2x_2 + y_1 - 5y_2 \\
 & \text{s.t.} \quad 3x_1 - x_2 - 2y_1 + y_2 \leq 4 \quad (1) \\
 & \quad \quad x_1 + x_2 \leq 1 \quad (2) \\
 & \quad \quad 0 \leq x_i \leq 1 \quad \forall i = 1, 2 \quad (3) \\
 & \quad \quad y_1 + y_2 \geq 1 \quad (4) \\
 & \quad \quad 0 \leq y_i \leq 1 \quad \forall i = 1, 2. \quad (5)
 \end{aligned}$$

This is a toy example of a structured linear program. Namely, constraints (2)-(3) define a simple polytope for x_1, x_2 , denoted as P . Constraints (4)-(5) define a simple polytope for y_1, y_2 denoted as Q . Constraint (1) couples all the variables x_i 's and y_i 's together. We will reformulate (S) using Dantzig-Wolfe decomposition.

- (a) Denote the extreme points of P as $\mathbf{x}^1, \mathbf{x}^2, \dots, \mathbf{x}^m$, and the extreme points of Q as $\mathbf{y}^1, \mathbf{y}^2, \dots, \mathbf{y}^k$. Write down the numerical values of all these extreme points in the format of vectors $\begin{bmatrix} a \\ b \end{bmatrix}$, specify the values of a, b for each extreme point.

- (b) Any point $\mathbf{x} \in P$ can be written as a convex combination the extreme points of P , i.e. $\mathbf{x} = \sum_{i=1}^m \lambda_i \mathbf{x}^i$, where λ_i 's are convex weights. Similarly, any point $\mathbf{y} \in Q$ can be written as $\mathbf{y} = \sum_{i=1}^k \mu_i \mathbf{y}^i$, where μ_i 's are convex weights. Reformulate the LP (S) in variables λ_i 's and μ_i 's. You need to write down explicitly the numerical values of all the coefficients in all constraints and the objective function of the new formulation. Hint: you need to use extreme point representations for both P and Q .

5. Two-Stage Stochastic Program

Consider a farmer Anne who specializes in raising wheat, corn, and sugar beets on her 500 acres of land. During the winter, she wants to decide how much land to devote to each crop.

Anne knows that at least 200 tons (T) of wheat and 240 T of corn are needed for cattle feed. These amounts can be raised on the farm or bought from a wholesaler. Any production in excess of the feeding requirement would be sold. Over the last decade, average selling prices have been \$170 and \$150 per ton of wheat and corn, respectively. The purchase prices are 40% more than this due to the wholesaler's margin and transportation costs.

Another profitable crop is sugar beet, which Anne expects to sell at \$36/T; however, the Agriculture Commission imposes a quota on sugar beet production. Any amount in excess of the quota can be sold only at \$10/T. Anne's quota for next year is 6000 T.

Based on past experience, Anne knows that the average yield on her land is roughly 2.5 T, 3 T, and 20 T per acre for wheat, corn, and sugar beets, respectively.

The planting costs for wheat, corn, and sugar beet are \$150/acre, \$230/acre, and \$260/acre, respectively.

The following table summarizes these data.

	Wheat	Corn	Sugar Beets
Unit Yield (T/acre)	2.5	3	20
Selling price (\$/T)	170	150	36 under 6000T 10 above 6000T
Purchase price (\$/T)	238	210	-
Planting cost (\$/acre)	150	230	260
Min requirement (T)	200	240	-
Total land: 500 acres			

- (a) Set up a deterministic linear program to help Anne make an optimal plan to maximize the profit for next year's farming. Use the following definition of variables.

x_1 = acres of land devoted to wheat,

x_2 = acres of land devoted to corn,

x_3 = acres of land devoted to sugar beets,

y_1 = tons of wheat purchased,

y_2 = tons of corn purchased,

w_1 = tons of wheat sold,

w_2 = tons of corn sold,

w_3 = tons of sugar beets sold at the favorable price,

w_4 = tons of sugar beets sold at the lower price.

- (b) According to Anne's experience, it is quite common that the yields of the three crops can significantly vary above or below the forecast values given in the above table. It is thus important to account for this uncertainty. Assume there are three scenarios for the yields, with 1/3 probability, the yields of all three crops will be 20% above the forecast values, with 1/3 probability, they will be equal to the forecast values, and with 1/3

probability they will be 20% below the forecast values. Formulate a two-stage stochastic linear program to maximize the average profit.

- (c) Solve the deterministic LP and the extended formulation of the two-stage stochastic LP in Xpress and compare the results. What can you say about the advantages of the stochastic LP versus and deterministic LP? (This exercise gives you a concrete example to study stochastic LP. In the final exam, you will not be asked to write Xpress code.)

6. Integer Programming Computation

Consider the integer programming problem:

$$\begin{array}{ll}\max & x_1 + 2x_2 \\ \text{s.t.} & -x_1 + x_2 \leq 1 \\ & 4x_1 + 3x_2 \leq 6 \\ & x_1, x_2 \geq 0, \text{ integer}\end{array}$$

- (a) What are the optimal cost and optimal solution of the integer programming problem? What are the optimal cost and optimal solution of the linear relaxation problem? You can use the grid provided on the next page to solve the problems graphically.
- (b) Write down linear inequalities that describe the integer hull.
- (c) Solve the problem by branch-and-bound. Draw the complete branch-and-bound tree. Solve the LP relaxations graphically using the grid provided on the next page.

- (d) Suppose we dualize the constraint $-x_1 + x_2 \leq 1$. Write down the Lagrangian relaxation problem explicitly as an optimization problem over the Lagrangian multiplier.

- (e) Write down the Lagrangian dual problem in the original space of x_1, x_2 , using the Theorem of Lagrangian Duality for Integer Optimization. [Hint: You need to consider the convex hull of some integer set.] Solve the Lagrangian dual problem graphically. What is the optimal solution in the x_1, x_2 space, and what is the optimal cost of the Lagrangian dual problem?

Solution: Using the Theorem of Lagrangian Duality for Integer Optimization, the Lagrangian dual problem can be written as

$$Z_{LD} = \max \quad x_1 + 2x_2 \tag{6}$$

$$\text{s.t.} \quad -x_1 + x_2 \leq 1 \tag{7}$$

$$2x_1 + x_2 \leq 2 \tag{8}$$

$$x_1 \geq 0, x_2 \geq 0 \tag{9}$$

where (8)-(9) describe the convex hull of the integer set $\{(x_1, x_2) : 4x_1 + 3x_2 \leq 6, x_1, x_2 \geq 0, \text{integer}\}$. It is easy to solve this Lagrangian dual problem graphically. The optimal solution is achieved at the intersection of $-x_1 + x_2 = 1$ and $2x_1 + x_2 = 2$, i.e. $x_1 = 1/3, x_2 = 4/3$, and the optimal cost of the Lagrangian dual is $Z_{LD} = 3$. Notice that the LP relaxation $Z_{LP} = 23/7 = 3.286$ and the IP's optimal cost $Z_{IP} = 2$. So we have $Z_{LP} > Z_{LD} > Z_{IP}$, i.e. Lagrangian relaxation gives a better bound than the LP relaxation.

7. One More Exercise on Integer Programming Computation

Consider the integer programming problem:

$$\begin{array}{ll}\min & -x_1 - 2x_2 \\ \text{s.t.} & -3x_1 + 4x_2 \leq 4 \\ & 5x_1 + 4x_2 \leq 10 \\ & x_1, x_2 \geq 0, \text{ integer}\end{array}$$

- Solve the LP relaxation graphically. What are the optimal cost and optimal solution of the linear programming relaxation?
- Solve the IP graphically. What are the optimal cost and optimal solution of the integer programming problem?
- What is the integer hull of the set of feasible integer solutions? Show the integer hull graphically. Then write down all the constraints that define the integer hull.
- Solve the problem by branch-and-bound. Solve the LP relaxations in the branch-and-bound process graphically.
- Derive a Chvatal-Gomory cut generated from the LP relaxation of the original IP problem.
- Suppose we dualize the constraint $-3x_1 + 4x_2 \leq 4$. Let the Lagrangian multiplier be y . Write down the Lagrangian relaxation problem. Specify the sign of the multiplier y .
- Write the optimal cost of the Lagrangian relaxation problem as a function of the multiplier y . Is this function a convex or concave piecewise linear function? Graph this function.
- Write down Lagrangian dual problem in the space of the Lagrangian multiplier y , i.e. the dual variable. Reformulate it as an linear programming problem in the space of y and one additional variable.
- Solve the above Lagrangian dual problem in the y -space graphically. What is the optimal cost of the Lagrangian dual problem? Compare it with the optimal objective values of LP relaxation and the original IP problem.
- Reformulate the Lagrangian dual problem in the x_1, x_2 space, using the Theorem of Primal Characterization of Lagrangian Duality for Integer Program. Solve the resulting Lagrangian dual problem graphically. What is the optimal solution x_1, x_2 ?

Solution of Problem 5

- Figure 1 shows the graphical solution, highlighted in a big green dot. With some basic algebra we can calculate that the optimal solution is $x_{LP}^* = (\frac{3}{4}, \frac{25}{16})$. And the value of this solution is $Z_{LP} = -3\frac{7}{8}$.
- The integer program has five feasible integer solutions: $S = \{(0, 0), (1, 0), (2, 0), (0, 1), (1, 1)\}$ shown by the purple points in Figure 2. By simple inspection it follows that $x_{IP}^* = (1, 1)$ is the optimal integer solution with optimal objective value of the integer program $Z_{IP}^* = -3$, as highlighted by the big green dot.

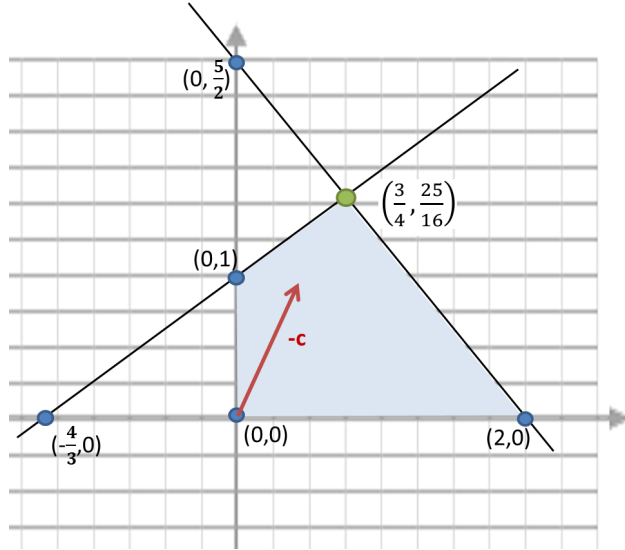


Figure 1: LP Relaxation

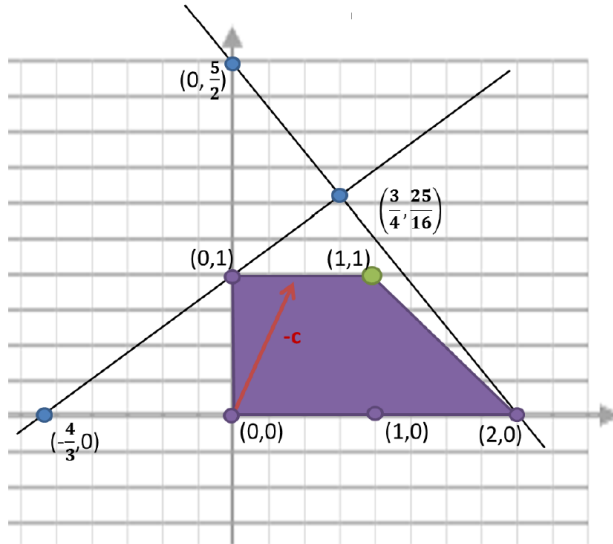


Figure 2: Integer hull, and IP solution

- (c) The integer hull of the IP is the convex hull of the set of feasible integer solutions $S = \{(0, 0), (1, 0), (2, 0), (0, 1), (1, 1)\}$. Figure 2 shows the graph of the integer hull represented by the purple area. The linear constraints describing the integer hull are the following:

$$x_1 + x_2 \leq 2, \quad x_2 \leq 1, \quad x_1 \geq 0, \quad x_2 \geq 0.$$

- (d) We begin with the LP relaxation solution above, since both variables are fractional we can branch on any one. We begin branching on x_2 , note that $1 \leq 3/2 \leq 2$. The first node, with the constraint $x_2 \geq 2$ can easily be seen to be infeasible on the original problem

graph, so we can prune this node by infeasibility. Now we focus our attention on the node with $x_2 \leq 1$. We solve this LP relaxation in Figure 3.

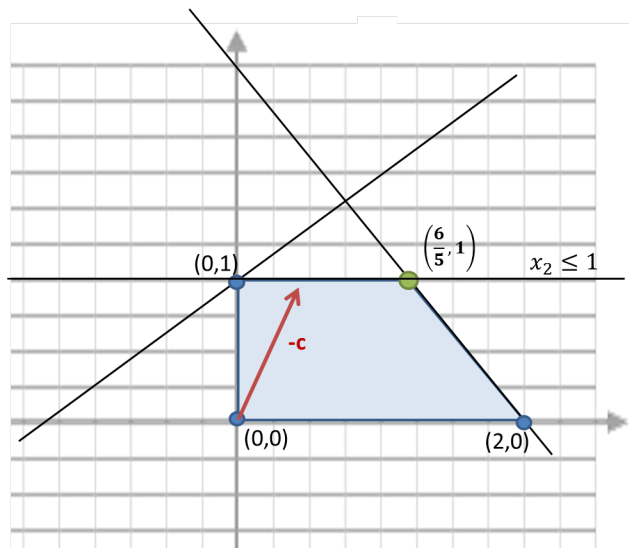


Figure 3: Branch and Bound solution, step 2.

We once again have a fractional solution with $x = (\frac{6}{5}, 1)$, and an optimal value $Z = -3\frac{1}{6}$. So now we will branch on x_1 . First we explore the node with the new constraint $x_1 \leq 1$. The result of this node is seen in Figure 4.

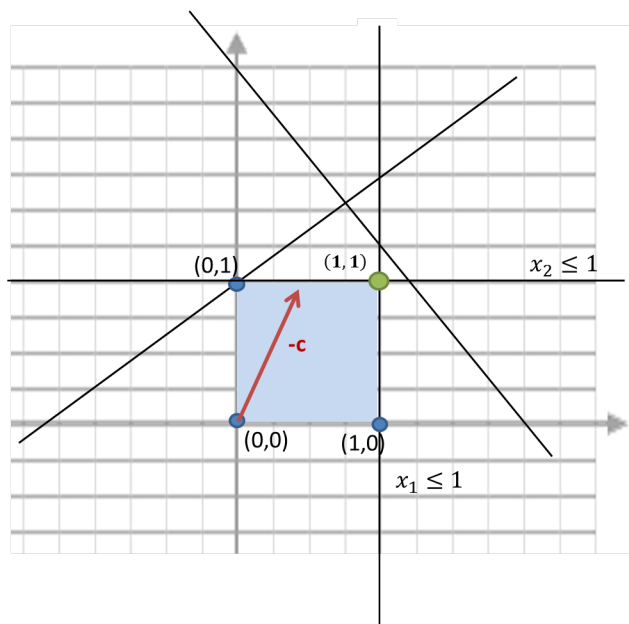


Figure 4: Branch and Bound solution, step 3.

The optimal solution to this node is $x^* = (1, 1)$ and $Z^* = -3$ (which we will see is the

optimum), this value becomes our first (and only) best upper bound. Finally we open the node with the constraint $x_1 \geq 2$, there is only one point feasible for this polyhedron, which is $x = (0, 2)$ with $Z = -2$, since this is no better than the previous point we prune this node by bound. Since all open nodes have been pruned we are done.

(e) First form the standard form problem:

$$\begin{aligned} \min \quad & -x_1 - 2x_2 \\ \text{s.t.} \quad & -3x_1 + 4x_2 + x_3 = 4 \\ & 5x_1 + 4x_2 + x_4 = 10 \\ & x_1, x_2, x_3, x_4 \geq 0, \text{ integer} \end{aligned}$$

Remember, for the LP Relaxation of the above problem, we had $x_{LP}^* = (\frac{3}{4}, \frac{25}{16}, 0, 0)$. Now let us add the two linear equalities. We get

$$2x_1 + 8x_2 + x_3 + x_4 = 14$$

Now divide by 8 to make the coefficient of x_2 to be 1,

$$\frac{1}{4}x_1 + x_2 + \frac{1}{8}x_3 + \frac{1}{8}x_4 = \frac{7}{4}$$

Applying round down to the coefficients and right-hand side, we get the following Chvatal-Gomory cut,

$$x_2 \leq 1$$

which is the same constraint we introduced in the branch and bound.

(f) The Lagrangian relaxation of the problem is

$$\begin{aligned} Z(y) = \min_{x_1, x_2} \quad & -x_1 - 2x_2 + y(-3x_1 + 4x_2 - 4) \\ \text{s.t.} \quad & 5x_1 + 4x_2 \leq 10 \\ & x_1, x_2 \geq 0, \text{ integer} \end{aligned}$$

where $y \geq 0$.

(g) We can solve the Lagrangian relaxation by observing that $Z(y)$ can be rewritten as:

$$\begin{aligned} Z(y) = \min_{x_1, x_2} \quad & -x_1 - 2x_2 + y(-3x_1 + 4x_2 - 4) \\ \text{s.t.} \quad & (x_1, x_2) \in \{(0, 0), (0, 1), (0, 2), (1, 0), (1, 1), (2, 0)\} \\ & = \min \{-4y, -2, 4y - 4, -7y - 1, -3y - 3, -10y - 2\} \end{aligned}$$

Figure 5 on the next page shows the plot of $Z(y)$ as a piecewise linear concave function of y .

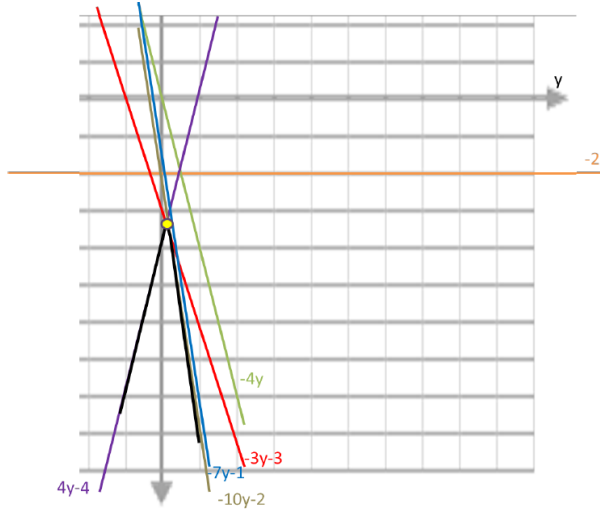


Figure 5: Function $Z(y)$.

(h) The Lagrangian dual problem is,

$$\begin{aligned} Z_{LD} = \max_y \quad & Z(y) \\ \text{s.t.} \quad & y \geq 0. \end{aligned}$$

Substituting the expression for $Z(y)$ obtained above, we have

$$\begin{aligned} Z_{LD} = \max_y \quad & \min \{-4y, -2, 4y - 4, -7y - 1, -3y - 3, -10y - 2\} \\ \text{s.t.} \quad & y \geq 0. \end{aligned}$$

Notice that this is not a linear program, having a concave piecewise linear objective function. We can reformulate the problem as a linear program as follows:

$$\begin{aligned} Z_{LD} = \max_{y,z} \quad & z \\ \text{s.t.} \quad & z \leq -4y \\ & z \leq -2 \\ & z \leq 4y - 4 \\ & z \leq -7y - 1 \\ & z \leq -3y - 3 \\ & z \leq -10y - 2 \\ & y \geq 0. \end{aligned}$$

(i) To solve this Lagrangian dual problem, we can use Figure 5 to find the maximum of $Z(y)$. As a result, the maximizer of $Z(y)$ for $y \geq 0$ is achieved at the intersection of $4y - 4 = -3y - 3$, which gives $y = 1/7$, and $Z_{LD} = 4/7 - 4 = -24/7$. Comparing the LP

relaxation $Z_{LP} = -3\frac{7}{8} = -3.875$, the Lagrangian dual $Z_{LD} = -3\frac{3}{7} \approx -3.428$, and the IP $Z_{IP} = -3$, we get:

$$Z_{LP} < Z_{LD} < Z_{IP}.$$

- (j) Using the Theorem of Primal Characterization of Lagrangian Duality for Integer Program, we can reformulate the Lagrangian dual in the x_1, x_2 space as

$$\begin{aligned} \min \quad & -x_1 - 2x_2 \\ \text{s.t.} \quad & -3x_1 + 4x_2 \leq 4 \\ & x_1 + x_2 \leq 2 \\ & x_1, x_2 \geq 0. \end{aligned}$$

Note that the second and third constraints describe the convex hull of $\{(x_1, x_2) : 5x_1 + 4x_2 \leq 10, x_1, x_2 \geq 0, \text{ integer}\}$. The optimal solution of the Lagrangian dual can be easily obtained by looking at the graph of the feasible region (omitted). The optimal solution is the intersection of $-3x_1 + 4x_2 = 4$ and $x_1 + x_2 = 2$, which is $x_1^* = \frac{4}{7}, x_2^* = \frac{10}{7}$ and $Z_{LD} = -\frac{24}{7}$.